

The empirical recurrence for OEIS A256731, recovered from its tail

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8 September 2026

Abstract

OEIS A256731 records “Empirical recurrence of order 78” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 368$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 81$. Its last coefficient is $c_{78} = 1361129467683753853853498429727072845824 \neq 0$, so no further tail satisfies anything shorter; any order-78 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A256731 is “Number of $(n+2) \times (4+2)$ 0..1 arrays with no 3×3 subblock diagonal sum zero and no antidiagonal sum three.”. It has offset 1 and begins

73984, 1648656, 30294016, 635846656, 12465275904, 249872016384, 4910514177024, 97652975330304, ...

The entry states:

Empirical recurrence of order 78 (see link above).

As of the “Last modified” line on the live entry (Jan 16 2022 01:30:51, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 368$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 368$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 78: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 78.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 736$ exact terms from $n = 3$ on, it returns order 78 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	31	c_{21}	13281994997211070464	c_{41}	−49539820264652928875298711142
c_2	−199	c_{22}	−4517845908890583040	c_{42}	−359055150937342195236915086622
c_3	−1683	c_{23}	−174563735410292555776	c_{43}	−677522417029199242863629146521
c_4	38100	c_{24}	−1130706829141803532288	c_{44}	−776996798506799269481959286374
c_5	−278840	c_{25}	−4804344074305835892736	c_{45}	−946160004283965592695749707038
c_6	1633904	c_{26}	−8980401005993178169344	c_{46}	4076162330057757710072460848660
c_7	−38552080	c_{27}	−76080211539832051073024	c_{47}	4010052938300891063286214147702
c_8	109914176	c_{28}	−524433122155426977153024	c_{48}	16091394738950409926291290914291
c_9	955966720	c_{29}	−1560842313208361402761216	c_{49}	25815931794726638015927382246948
c_{10}	355537280	c_{30}	−7708818660531970240413696	c_{50}	−4988838554034700030529752272994
c_{11}	100416226432	c_{31}	−34629911214197221477056512	c_{51}	−30101253997353396825790866358953
c_{12}	−580233449472	c_{32}	−79441249342502171299545088	c_{52}	−49888009366708301289639717065300
c_{13}	5640801163776	c_{33}	−201767494580785361629216768	c_{53}	−1851951342201018401755564848709
c_{14}	−22212029773824	c_{34}	421220883348064014570618880	c_{54}	190885798928191420586185639790746
c_{15}	−68409253369856	c_{35}	7821623351202131490505752576	c_{55}	451953515964216308574333751929706
c_{16}	−1967703474573312	c_{36}	38154877496118787468406292480	c_{56}	174714525368540519072180922835428
c_{17}	−14603201596903424	c_{37}	133015785266497791809653047296	c_{57}	−105601730289061476355169599183954
c_{18}	10870842634588160	c_{38}	339255278863255392518058016768	c_{58}	−231802459406201015196321542192748
c_{19}	488673127417925632	c_{39}	937634727360168809134773239808	c_{59}	−187211279009430395002672220277351
c_{20}	3748618121045721088	c_{40}	2044624095306089147644669591552	c_{60}	1408846506805509958072666439237710

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 81$, and it is the recurrence OEIS A256731 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{78} - c_1 x^{77} - \dots - c_{78}$. Its constant term is $-c_{78} = -1361129467683753853853498429727072845824$,

which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 78 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 78, so $\deg q = 78$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 78, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 78, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1361129467683753853853498429727072845824.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-78 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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